

PocketSmart AI

Project Design Phase

1. Design Overview

PocketSmart AI is designed as a modular web application. The system receives user planning requirements, processes budget and preferences, generates recommendations using AI when available, and displays the results through a responsive web interface.

2. System Architecture

- **Presentation Layer:** HTML, CSS, JavaScript and Jinja2
- **Application Layer:** FastAPI routes
- **Service Layer:** Gemini AI and recommendation service
- **Data Layer:** SQLite database

3. System Flow

User → Login/Register → Select Planner → Enter Budget & Preferences → FastAPI → Recommendation Service → Gemini AI / Fallback → Results

4. Main Modules

- Authentication
- Home Planner
- Party Planner
- Jewelry Planner
- AI Recommendation Service
- Fallback Recommendation Service
- Database

5. Use Case Design

Primary Actor: User

- Register account
- Login
- Select planner
- Enter budget and preferences
- Request recommendations

- View recommendation results
- View estimated cost
- Logout

6. Data Flow Design

User Input → FastAPI → Validation → Recommendation Service → Gemini AI / Fallback Service → Recommendation Response → Web Interface

7. Database Design

Users Table

Field	Type	Description
id	INTEGER	Primary key
name	TEXT	User name
email	TEXT	Unique email
password	TEXT	User password data

8. UI Design

- Navigation bar
- Homepage
- Planner forms
- Recommendation result cards
- Login and Register pages
- Responsive design

9. Technology Architecture

- **Frontend:** HTML, CSS, JavaScript, Jinja2
- **Backend:** Python, FastAPI
- **Database:** SQLite
- **AI:** Google Gemini API
- **Development Tool:** Visual Studio Code
- **Version Control:** GitHub

10. Design Summary

The design separates the user interface, API routes, business services and database operations into modular components. This makes PocketSmart AI easier to develop, test, maintain and extend.